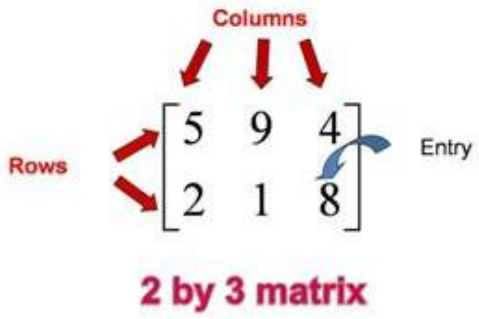
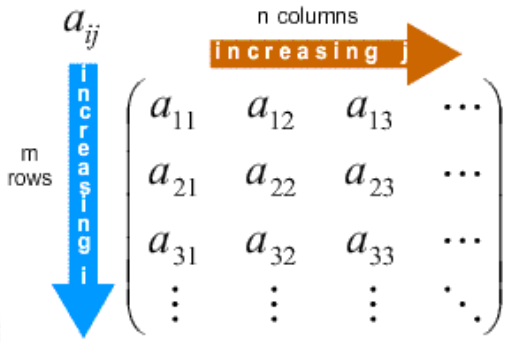
# Introduction to Matrices （矩陣簡介）

In mathematics, an $m \times n$ **matrix** (矩陣) is a rectangular array of m rows n columns. The elements in the matrix can be numbers, symbols or mathematical expressions.



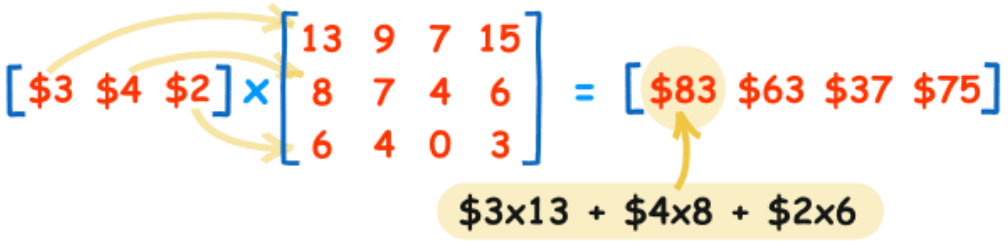




Matrices are a common tool in linear algebraic mathematics, as well as in applied mathematics such as statistical analysis. In physics, matrices are used in electrical circuits, forces, optics, and quantum physics; and in computer science, matrices are needed for three-dimensional animation. Simple applications of matrices are as follows:

e.g. A fruit store sells the following fruits: apple ($3 each), banana ($4 each) and orange ($2 each). The sales in 4 days are shown in the following table:

| | Day 1 | Day 2 | Day 3 | Day 4 |
|---|---|---|---|---|
| Apple | 13 | 9 | 7 | 15 |
| Banana | 8 | 7 | 4 | 6 |
| Orange | 6 | 4 | 0 | 3 |

If the data above is represented by matrics, it will be easier to calculate the total sales amount as follows:




The sales in day 1 is $83, day 2 is $63, etc.


***Not for sale***

# Concepts of Matrix (矩陣之概念)

**Definition: Matrix (矩陣之定義)**

A rectangular array of $m \times n$ numbers arranged in the form

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

is called an $m \times n$ matrix ($m \times n$ 階矩陣).

**Remarks:**

**1.** The numbers $a_{ij}$ $(i = 1, 2, ..., m;\ j = 1, 2, ..., n)$ are called ***elements* (元素)**.

**2.** $m \times n$ is called the ***order* (階)** of the matrix.

**3.** If $m = n$, then the matrix is a ***square matrix of order n* (*n* 階方陣)**. The diagonal containing $a_{11}, a_{22}, ..., a_{nn}$ is called the ***principal diagonal* (主對角線)**.

**Examples:**

- $\begin{pmatrix} 1 & 2 \\ 3 & -4 \end{pmatrix}$ is a square matrix of order 2, with principal diagonal {1, –4}
- $\begin{pmatrix} 3 & 1 & 0 \\ 0 & 0 & 2 \\ 5 & 4 & -1 \end{pmatrix}$ is a square matrix with order 3, principal diagonal {3, 0 –1}

**4.** When $m = 1,\ A = (a_{11}, a_{12}, ..., a_{1n})$ is called a ***row vector* (行向量)**.

**5.** When $n = 1,\ A = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}$ is called a ***column vector* (列向量)**.


***Not for sale***



# Concepts of Matrix

**Remarks:**

**6.** If all the elements are zero, the matrix is called a ***zero matrix* (零矩陣)** and it is usually denoted by **O**.

**Examples:**

- $\mathbf{O} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is a $2\times 2$ zero matrix
- $\mathbf{O} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a $2\times 3$ zero matrix.

**7.** A square matrix whose elements except the principal diagonal are all zero is called a ***diagonal matrix* (對角矩陣)**.

**Examples:**

$\begin{pmatrix} 1 & 0 \\ 0 & -4 \end{pmatrix}$ and $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ are diagonal matrices.

**8.** A diagonal matrix of order $n$ whose elements in the principal diagonal are all 1 is called an ***identity matrix*** or a ***unit matrix* (單位方陣)** and is denoted by $I_n$ or simply by $I$.

**Examples:**

- $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is an identity matrix of order 2
- $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is an identity matrix of order 3.


***Not for sale***

# Arithmetic of Matrices (矩陣之運算)

**Definition: Equality of Two Matrices (矩陣相等之定義)**

Two matrices $A=\left[a_{ij}\right]_{m\times n}$ and $B=\left[b_{ij}\right]_{m\times n}$ are ***equal*** if and only if they are of the same order and their corresponding elements are equal,

i.e. $a_{ij}=b_{ij}$ for all $i$ and $j$.

**Example:**

If $\begin{pmatrix} a & b \\ c & d \end{pmatrix}=\begin{pmatrix} 1 & 3 \\ 6 & 8 \end{pmatrix}$, then $a=1, b=3, c=6$ and $d=8$

**Definition: Addition of Matrices (矩陣加法之定義)**

Let $A=\left[a_{ij}\right]_{m\times n}$ and $B=\left[b_{ij}\right]_{m\times n}$ be two $m\times n$ matrices. The sum of $A$ and $B$, denoted by $A+B$, is a matrix of order $m\times n$ and its elements is the sum of the corresponding elements of $A$ and $B$,

i.e., $A+B=\left[a_{ij}+b_{ij}\right]_{m\times n}$

**Examples:**

- $\begin{pmatrix} 3 & 2 & 5 \\ 4 & 3 & 2 \end{pmatrix}+\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}=\begin{pmatrix} 4 & 3 & 6 \\ 6 & 5 & 4 \end{pmatrix}$
- $\begin{pmatrix} 2 & 1 \\ 3 & 0 \\ -1 & 4 \end{pmatrix}+\begin{pmatrix} 2 & 3 & -1 \\ 1 & 0 & 4 \end{pmatrix}$ is not defined.
- $\begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix}+5$ is not defined.


***Not for sale***

# Arithmetic of Matrices

**Definition: Negative Matrices (負矩陣之定義)**

Let $A=\left[a_{ij}\right]_{m\times n}$ be an $m\times n$ matrix. The ***negative*** of $A$, denoted by $-A$, is a matrix whose elements are the negative of the corresponding elements of $A$.

i.e. $-A=\left[-a_{ij}\right]_{m\times n}$

**Example:**

If $A=\begin{pmatrix}5 & 0\\ -1 & 4\end{pmatrix}$, then $-A=\begin{pmatrix}-5 & 0\\ 1 & -4\end{pmatrix}$

**Definition: Subtraction of Matrices (矩陣減法之定義)**

The ***difference*** of two matrices $A$ and $B$ is defined as $A-B=A+(-B)$. That is,

$A-B=\left[a_{ij}-b_{ij}\right]_{m\times n}$

**Example:**

$$\begin{pmatrix}-5 & 8\\ 2 & 1\end{pmatrix}-\begin{pmatrix}0 & 2\\ -4 & 6\end{pmatrix}=\begin{pmatrix}-5 & 6\\ 6 & -5\end{pmatrix}$$



**Theorem: Properties of Matrix Addition (矩陣加法性質)**

Let $A$, $B$, $C$ be $m\times n$ matrices and 0 be the corresponding $m\times n$ zero matrix. Then

**(1)** $A+B=B+A$ (Commutative Law)

**(2)** $(A+B)+C=A+(B+C)$ (Associative Law)

**(3)** $A+0=0+A=A$

**(4)** $A+(-A)=(-A)+A=0$



# Arithmetic of Matrices

In mathematics, scalar (純量) is a quantity that has only a magnitude but no direction (a quantity with direction is called a vector, which will be introduced in a later topic). Pure quantities can be any real numbers (positive, negative, fractional, decimal...) and the operation of multiplying scalar by a matrix is very simple and straightforward:

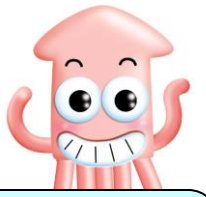
**Definition: Scalar Multiplication of Matrices (矩陣純量乘法之定義)**

Let $A=\left[a_{ij}\right]_{m\times n}$ be an $m\times n$ matrix and $k$ a scalar. Then $kA$ is defined to be a matrix obtained by multiplying each element of $A$ by $k$,

i.e. $kA=\left[ka_{ij}\right]_{m\times n}$

**Examples:**

- If $A=\begin{pmatrix}3 & -2 & 1\\ -5 & 6 & 7\end{pmatrix}$, then $-3A=\begin{pmatrix}-9 & 6 & -3\\ 15 & -18 & -21\end{pmatrix}$.
- $\begin{pmatrix}-2 & 4 & -6\\ -4 & 6 & -8\end{pmatrix}=2\begin{pmatrix}-1 & 2 & -3\\ -2 & 3 & -4\end{pmatrix}=-2\begin{pmatrix}1 & -2 & 3\\ 2 & -3 & 4\end{pmatrix}$



**Theorem: Properties of Scalar Multiplication (純量乘法性質)**

Let $A$, $B$ be two $m\times n$ matrices and $h$, $k$ be two scalars. Then

**(1)** $k(A+B)=kA+kB$

**(2)** $(h+k)A=hA+kA$

**(3)** $(hk)A=h(kA)=k(hA)$



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# Transpose of Matrices (轉置矩陣)

**Definition: Transpose of Matrix (轉置矩陣之定義)**

Let $A=\left[a_{ij}\right]_{m\times n}$ be an $m\times n$ matrix. The ***transpose*** of $A$, denoted by $A^t$ or $A^T$, is the $n\times m$ matrix obtained by interchanging the rows and columns in $A$, i.e. the $i$ th row of $A$ becomes the $i$ th column of $A^T$.

**Examples:**

- If $A=\begin{pmatrix}3 & -2 & 4\\ 1 & 0 & -3\end{pmatrix}$, then $A^T=\begin{pmatrix}3 & 1\\ -2 & 0\\ 4 & -3\end{pmatrix}$.
- If $B=\begin{pmatrix}5 & 0\\ -1 & 4\end{pmatrix}$, then $B^T=\begin{pmatrix}5 & -1\\ 0 & 4\end{pmatrix}$
- If $D=\begin{pmatrix}3 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & -1\end{pmatrix}$, then $D^T=D=\begin{pmatrix}3 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & -1\end{pmatrix}$
- If $I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, then $I^T=I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$.

---

**Remarks:**

**1.** A square matrix satisfying the condition that $A^T=A$ is called a ***symmetric matrix*** (對稱方陣).

**Examples:**

- $\begin{pmatrix}1 & 3\\ 3 & 5\end{pmatrix}$ and $\begin{pmatrix}3 & 1 & 2\\ 1 & 0 & 4\\ 2 & 4 & -1\end{pmatrix}$ are symmetric matrix.
- Any ***diagonal matrix*** and ***identity matrix*** must be symmetric matrix.


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# Transpose of Matrices

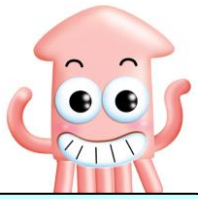
**Remarks:**

**2.** A square matrix satisfying the condition that $A^T = -A$ is called a ***skew-symmetric matrix*** or an ***anti-symmetric matrix*** **(反對稱方陣)**. Note that elements of the principal diagonal of a skew-symmetric matrix are zero.

**Example:**

$\begin{pmatrix} 0 & 3 & -1 \\ -3 & 0 & 5 \\ 1 & -5 & 0 \end{pmatrix}$ is a skew-symmetric matrix.

**3.** The zero matrix **0** is both symmetric and skew-symmetric,
i.e. $0^T = 0$ and $0^T = -0$.



**Theorem: Properties of Transpose of Matrix (轉置矩陣性質)**

Let $A$, $B$ be two $m \times n$ matrices. Then

**(1)** $(A^T)^T = A$

**(2)** $(A+B)^T = A^T + B^T$

**(3)** $(A-B)^T = A^T - B^T$

**More Example:**

Let $A$ be any square real matrix. If $S = A + A^T$ .Show that $S^T = S$

**Solution**
$$\begin{aligned} S^T &= (A + A^T)^T \\ &= A^T + (A^T)^T \\ &= A^T + A \\ &= S \end{aligned}$$



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# Multiplication of Matrices (矩陣之乘法)

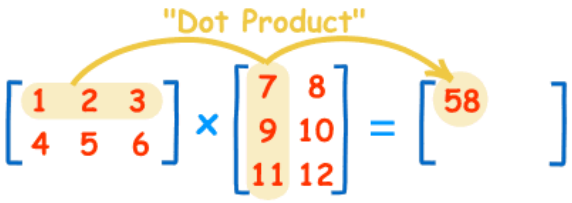
**Definition: Multiplication of Matrix (矩陣乘法之定義)**

Let $A=\left[a_{ij}\right]_{m\times r}$, $B=\left[b_{ij}\right]_{r\times n}$. The ***product*** $AB$ of $A$ and $B$ is a $m\times n$ matrix $C=\left[c_{ij}\right]_{m\times n}$ such that $c_{ij}=\sum_{k=1}^{r}a_{ik}b_{kj}=a_{i1}b_{1j}+a_{i2}b_{2j}+\cdots+a_{ir}b_{rj}$

**e.g.** If $A=\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{pmatrix}$ and $B=\begin{pmatrix}7 & 8\\ 9 & 10\\ 11 & 12\end{pmatrix}$, find $AB$.

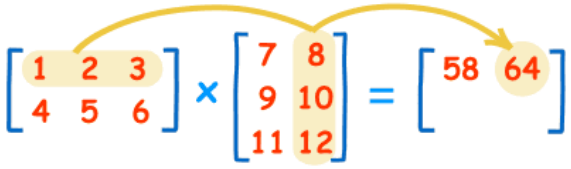
**Solution:**

首先將 ***A*** 的第 1 列 (橫) 與 ***B*** 的第 1 行 (直) 的數字相乘再加，將結果放於第 1 列、第 1 行，即是：



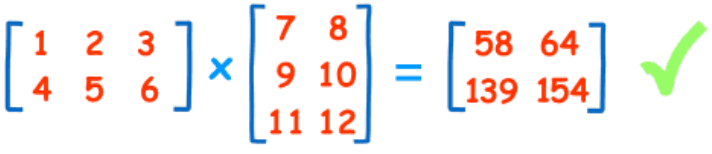

$$1\times7+2\times9+3\times11=58$$

然後將 ***A*** 的第 1 列 (橫) 與 ***B*** 的第 2 行 (直) 的數字相乘再加，將結果放於第 1 列、第 2 行，即是：




$$1\times8+2\times10+3\times12=64$$

如此類推，就能計算出 ***AB*** 的積：





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# Multiplication of Matrices

**Example 1:**

Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B=\begin{pmatrix}0 & -2 & 4\\ -1 & 3 & -5\end{pmatrix}$.

$$AB=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}0 & -2 & 4\\ -1 & 3 & -5\end{pmatrix}$$
$$=\begin{pmatrix}1\times 0+2\times(-1) & 1\times(-2)+2\times 3 & 1\times 4+2\times(-5)\\ 3\times 0+4\times(-1) & 3\times(-2)+4\times 3 & 3\times 4+4\times(-5)\end{pmatrix}$$
$$=\begin{pmatrix}-2 & 4 & -6\\ -4 & 6 & -8\end{pmatrix}$$
$$=2\begin{pmatrix}-1 & 2 & -3\\ -2 & 3 & -4\end{pmatrix}$$

However $BA$ is not defined (兩個矩陣大小不能配合，因此未能相乘).

**Example 2:**

Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B=\begin{pmatrix}5 & 7\\ 6 & 8\end{pmatrix}$.

$$AB=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}5 & 7\\ 6 & 8\end{pmatrix}$$
$$=\begin{pmatrix}1\times 5+2\times 6 & 1\times 7+2\times 8\\ 3\times 5+4\times 6 & 3\times 7+4\times 8\end{pmatrix}$$
$$=\begin{pmatrix}17 & 23\\ 39 & 53\end{pmatrix}$$

$$BA=\begin{pmatrix}5 & 7\\ 6 & 8\end{pmatrix}\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$$
$$=\begin{pmatrix}5\times 1+7\times 3 & 5\times 2+7\times 4\\ 6\times 1+8\times 3 & 6\times 2+8\times 4\end{pmatrix}$$
$$=\begin{pmatrix}26 & 38\\ 30 & 44\end{pmatrix}$$



**注意!!!** 在大多數的情況下，矩陣乘法是不可交換的（即 $AB\neq BA$ ），這與以往我們所學之乘法特性有很大出入，敬請留意!!

$$\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\times\begin{bmatrix}2 & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}4 & 4\\ 10 & 8\end{bmatrix}$$

$$\begin{bmatrix}2 & 0\\ 1 & 2\end{bmatrix}\times\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}=\begin{bmatrix}2 & 4\\ 7 & 10\end{bmatrix}$$


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# Multiplication of Matrices

*[P.Math 1989 Paper II Q.1(a)]*

Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 2 \\ -2 & 3 & -1 \end{pmatrix}$.

Find $AB^T$ and $B^T A$, where $B^T$ denote the transpose of $B$. ***(2 marks)***


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# Multiplication of Matrices (2)

**Theorem: Properties of Matrix Multiplication (矩陣乘法之性質)**

Let $k$ be a scalar and $A$, $B$, $C$ be matrices such that the following matrix products are defined. Then

**(1)** $(AB)C = A(BC)$ (Associative Law)

**(2)** $A(B+C) = AB + AC$ (Distributive Law)

**(3)** $(A+B)C = AC + BC$ (Distributive Law)

**(4)** $k(AB) = (kA)B = A(kB)$

**(5)** $(AB)^T = B^T A^T$ **(Transpose of the Product of Matrix)**

**(6)** $A0 = 0A = 0$

**(7)** $IA = AI = A$

**Remarks:**

**1.** Matrix multiplication is not commutative, i.e. $AB \neq BA$ in general.
Therefore most of the laws from algebra CANNOT be applied, for example:

- $$\begin{aligned}(AB)^2 &= (AB)(AB)\\ &= ABAB\\ &\neq A^2B^2\end{aligned}$$

- $$\begin{aligned}(A+B)(A-B) &= A^2 - AB + BA - B^2\\ &\neq A^2 - B^2\end{aligned}$$

- $$\begin{aligned}(A+B)^2 &= A^2 + AB + BA + B^2\\ &\neq A^2 + 2AB + B^2\end{aligned}$$

**2.** However, if $AB = BA$, the general rules of multiplication in algebra can also be applied in matrix multiplication.

**Example:**

If $AB = BA$, we have

$$(AB)^n = A^n B^n$$

$$(A+B)(A-B) = A^2 - B^2$$

$$(A \pm B)^2 = A^2 \pm 2AB + B^2$$

$$(A+B)^n = \sum_{r=0}^{n} C_r^n A^{n-r} B^r \qquad \text{(where } A^0 = I\text{)}$$



# Multiplication of Matrices (2)

**Remarks:**

**3.** A non-zero matrix may be a zero divisor,
i.e. even if $A \neq 0,\ B \neq 0$, we may still have $AB = 0$.

**Example:**

$$A=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix},\ B=\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix},\ AB=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}.$$

Hence,

(1) If $(A-k_1I)(A-k_2I)=0$, the result that

$$A=k_1I \quad \text{or} \quad A=k_2I$$

may **NOT** be true.

(2) The cancellation law is **NOT** true also,
i.e. $A \neq 0$ and $AB=AC \ \not\Rightarrow\ B=C$

The proof of this result is as follows:

$AB = AC$

$\Rightarrow AB - AC = O$

$\Rightarrow A(B-C) = O$

When $A \neq 0,\ B-C$ may be non-zero, and so $B$ and $C$ may not equal.



**注意!!!** 綜合來說，矩陣乘法與代數乘法有極大出入，例如

1. $AB \neq BA$
2. $(AB)^n \neq A^nB^n$
3. $AB = AC$ 不代表 $B = C$
4. $AB = 0$ 不代表 $A = 0$ 或 $B = 0$

這與以往我們所學之乘法特性有很大出入，敬請留意!!

# Multiplication of Matrices (2)

***[P.Math 2006 Paper I Q.8]***

Let $M = \begin{pmatrix} m & -m \\ m & m \end{pmatrix}$, where $m > 0$.

(a) Evaluate $M^2$. ***(1 mark)***

(b) Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a non-zero real matrix such that $MX = XM$.

Prove that $c = -b$ and $d = a$. ***(3 marks)***


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# $n^{th}$ Power of Square Matrix (方陣之冪)

由於矩陣之乘法複雜，因此若需要計算矩陣 $A$ 之冪 (次方) 例如 $A^2, A^3, \ldots$ 以至 $A^n$ 時，可以利用以下方法：

**Method 1:** 計算 $A^2, A^3, \ldots$ 並尋中箇中規律，從而推測 $A^n$

**Method 2:** 運用數學歸納法 (Mathematical Induction)

**Example 1:** Let $A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$. Find $A^2$ and $A^3$. Hence find $A^n$.

**Solution**

$$A^2 = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} = \begin{pmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & c^2 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & c^2 \end{pmatrix}\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} = \begin{pmatrix} a^3 & 0 & 0 \\ 0 & b^3 & 0 \\ 0 & 0 & c^3 \end{pmatrix}$$

$$\therefore \quad A^n = \begin{pmatrix} a^n & 0 & 0 \\ 0 & b^n & 0 \\ 0 & 0 & c^n \end{pmatrix}$$

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註: 從上述例子可見，對角方陣 (Diagonal Matrix) 之冪是較容易計算的。所以當同學於下一課題學會乘法反矩陣 (Inverse Matrix，A 乘法反矩陣會以 $A^{-1}$ 表示 )，就可以用 **Method 3: 矩陣對角化 (Diagonalizable matrix)** 去計算大多數矩陣的冪：

若 $X = ADA^{-1}$ (其中 $D$ 是對角方陣)

則 $X^n = (ADA^{-1})(ADA^{-1})(ADA^{-1}) \ldots (ADA^{-1})$

$= ADA^{-1}ADA^{-1}ADA^{-1} \ldots ADA^{-1}$

$= ADDD \ldots DA^{-1}$ (因為 $A^{-1}A = I$)

$= AD^nA^{-1}$


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# $n^{\text{th}}$ Power of Square Matrix

**Example 2:** Let $n$ be a positive integer.

Given that $A=\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}$ and $B=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}$.

(a) Find $A^n$

(b) Find $B^n$.

**Solution**

(a)

$$A^2=\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}^2=\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}=\begin{pmatrix}4 & 20\\ 0 & 0\end{pmatrix}=\begin{pmatrix}2^2 & 2(10)\\ 0 & 0\end{pmatrix}$$

$$A^3=\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}^3=\begin{pmatrix}4 & 20\\ 0 & 0\end{pmatrix}\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}=\begin{pmatrix}8 & 40\\ 0 & 0\end{pmatrix}=\begin{pmatrix}2^3 & 2^2(10)\\ 0 & 0\end{pmatrix}$$

$$\vdots$$

$$A^n=\begin{pmatrix}2 & 10\\ 0 & 0\end{pmatrix}^n=\begin{pmatrix}2^n & 2^{n-1}(10)\\ 0 & 0\end{pmatrix}$$

(b)

$$B^2=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}^2=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}$$

$$B^3=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}^3=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}$$

$$\vdots$$

$$B^n=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}^n=\begin{pmatrix}0 & -5\\ 0 & 1\end{pmatrix}$$


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# $n^{\text{th}}$ Power of Square Matrix

***[M2 2016 Q.8 Modified]***

(a) Define $A=\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}$. If $n$ is a positive integer, evaluate

(i) $A^2$,

(ii) $A^n$.

(b) Given that $\sum_{k=0}^{n-1} 2^k = 2^n - 1$, evaluate $\begin{pmatrix}1 & 0\\ 1 & 2\end{pmatrix}^n$. ***(6 marks)***


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# M.I. × $n^{\text{th}}$ Power of Square Matrix



涉及矩陣的證明題目中亦有不少是要用到數學歸納法 **(Mathematical Induction)**，敬請留意 ~

**Example 3:** Let $A=\begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix}$, where $a, b \in \mathbf{R}$ and $a \neq b$.

Prove that for all positive integers $n$, $A^n=\begin{pmatrix} a^n & \dfrac{a^n-b^n}{a-b} \\ 0 & b^n \end{pmatrix}$.

**Solution**

When $n=1$, the proposition is true obviously.

**Step 1:** 證明 **P(1)** 成立

Assume the proposition is true for $n=k$,

i.e. $A^k=\begin{pmatrix} a^k & \dfrac{a^k-b^k}{a-b} \\ 0 & b^k \end{pmatrix}$

**Step 2:** 假設對任意一個正整數 *k*, **P(*k*)** 成立

When $n=k+1$,

**Step 3:** 證明 **P(*k*+1)** 會成立

$$A^{k+1}=A^k \times A$$

$$=\begin{pmatrix} a^k & \dfrac{a^k-b^k}{a-b} \\ 0 & b^k \end{pmatrix}\begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix}$$

$$=\begin{pmatrix} a^{k+1} & a^k+b\left(\dfrac{a^k-b^k}{a-b}\right) \\ 0 & b^{k+1} \end{pmatrix}$$

$$=\begin{pmatrix} a^{k+1} & \dfrac{a^{k+1}-b^{k+1}}{a-b} \\ 0 & b^{k+1} \end{pmatrix}$$

Hence, the proposition is also true for $n=k+1$.

**Step 4:** 寫下結論。

By M.I., the proposition is true for all positive integers *n*.

# M.I. × $n^{th}$ Power of Square Matrix

***[M2 2014 Q.7]***

Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

Prove, by mathematical induction, that for all positive integers $n$, $A^{n+1} = 2^n A$.

***(6 marks)***


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# HKAL Questions (Level 2)

1. ***[P.Math 1995 Paper I Q.1]***

Let $A=\begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix}$, where $a, b \in \mathbf{R}$ and $a \neq b$.

(a) Prove that for all positive integers $n$, $A^n = \begin{pmatrix} a^n & \dfrac{a^n - b^n}{a-b} \\ 0 & b^n \end{pmatrix}$.

(b) Hence, or otherwise, evaluate $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}^{95}$. ***(6 marks)***


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# HKAL Questions (Level 3)

2. ***[P.Math 2000 Paper I Q.1]***

Let $M = \begin{pmatrix} 1 & 0 & 0 \\ \lambda & b & a \\ \mu & c & -b \end{pmatrix}$ where $b^2 + ac = 1$. Show by mathematical induction that

$$M^{2n} = \begin{pmatrix} 1 & 0 & 0 \\ n\left[\lambda(1+b)+\mu a\right] & 1 & 0 \\ n\left[\lambda c+\mu(1-b)\right] & 0 & 1 \end{pmatrix}$$ for all positive integers *n*.

Hence or otherwise, evaluate $\begin{pmatrix} 1 & 0 & 0 \\ -2 & 3 & 2 \\ 1 & -4 & -3 \end{pmatrix}^{2000}$. ***(5 marks)***


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# HKAL Questions (Level 4)

3. ***[P.Math 2004 Paper I Q.8]***

(a) Let $X$ and $Y$ be two square matrices such that $XY = YX$. Prove that

$$(X+Y)^n = \sum_{r=0}^{n} C_r^n X^{n-r} Y^r \quad \text{for } n = 3, 4, 5, \ldots$$

[Note: For any square matrix $A$, define $A^0 = I$ ]

(b) By using (a) and considering $\begin{pmatrix} 0 & 2 & 4 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$, find $\begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}^{100}$.


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# HKAL Questions (Level 4)


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# HKAL Questions (Level 4)

4. ***[P.Math 1999 Paper I Q.9]***

(a) Let $A$ and $B$ be two square matrices of the same order. If $AB = BA = 0$, show that $(A+B)^n = A^n + B^n$ for any positive integer $n$.

***(4 marks)***

(b) Let $A = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$ where $a$, $b$ are not both zero.

If $B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, show that $AB = BA = 0$ if and only if $p = r = 0$ and $aq + bs = 0$. ***(4 marks)***

(c) Let $C = \begin{pmatrix} x & y \\ 0 & z \end{pmatrix}$ where $x$, $z$ are non-zero and distinct. Find non-zero matrices $D$ and $E$ such that $C = D + E$ and $DE = ED = 0$.

***(3 marks)***

(d) Evaluate $\begin{pmatrix} 2 & 5 \\ 0 & 1 \end{pmatrix}^{99}$. ***(4 marks)***

# HKAL Questions (Level 4)


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# HKAL Questions (Level 4)

5. ***[P.Math 2004 Paper I Q.8]***

Let $A=\begin{pmatrix}\alpha-k & \alpha-\beta-k\\ k & \beta+k\end{pmatrix}$, where $\alpha,\ \beta,\ k\in\mathbf{R}$ with $\alpha\neq\beta$.

Define $X=\dfrac{1}{\alpha-\beta}(A-\beta I)$ and $Y=\dfrac{1}{\beta-\alpha}(A-\alpha I)$, where $I$ is the $2\times2$ identity matrix.

(a) Evaluate $XY$, $YX$, $X+Y$, $X^2$ and $Y^2$. ***(4 marks)***

(b) Prove that $A^n=\alpha^n X+\beta^n Y$ for all positive integers $n$. ***(4 marks)***

(c) Evaluate $\begin{pmatrix}5 & 4\\ 2 & 3\end{pmatrix}^{2004}$. ***(4 marks)***


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# HKAL Questions (Level 4)


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# HKDSE Questions (Level 4)

6. ***[M2 2015 Q.11]***

(a) Let $\lambda$ and $\mu$ be real numbers such that $\mu - \lambda \neq 2$. Denote the $2 \times 2$ identity matrix by $I$.

Define $A = \dfrac{1}{\lambda - \mu + 2}(I - \mu I + M)$ and $B = \dfrac{1}{\lambda - \mu + 2}(I + \lambda I - M)$,

where $M = \begin{pmatrix} \lambda & 1 \\ \lambda - \mu + 1 & \mu \end{pmatrix}$.

(i) Evaluate $AB$, $BA$ and $A + B$.

(ii) Prove that $A^2 = A$ and $B^2 = B$.

(iii) Prove that $M^n = (\lambda + 1)^n A + (\mu - 1)^n B$ for all positive integers $n$.

***(8 marks)***

(b) Using (a) or otherwise, evaluate $\begin{pmatrix} 4 & 2 \\ 0 & 6 \end{pmatrix}^{315}$. ***(4 marks)***

# HKDSE Questions (Level 4)


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# HKAL Questions (Level 4)

7. ***[P.Math 2008 Paper I Q.8]***

Denote the $2 \times 2$ identity matrix by $I$.

(a) Let $M = \begin{pmatrix} p & q \\ r & -p \end{pmatrix}$, where $p$, $q$ and $r$ are real numbers.

(i) Evaluate $M^2$.

(ii) Let $s$ be a real number. Prove that if $p^2 + qr = 1$, then for any positive integers $n$,

$$(M + sI)^{2n} = \frac{(s+1)^{2n} - (s-1)^{2n}}{2} M + \frac{(s+1)^{2n} + (s-1)^{2n}}{2} I .$$

***(7 marks)***

(b) Let $A = \begin{pmatrix} 6 & 3 \\ -1 & 2 \end{pmatrix}$.

(i) Express $A$ in the form $\begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} + \mu I$, where $\alpha, \beta, \gamma, \mu$ are real.

(ii) Evaluate $A^{800}$. ***(5 marks)***


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# HKAL Questions (Level 4)



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